

SigLIP2 for aerial fire risk classification

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Abstract

We examine the transfer of a pretrained SigLIP2 image encoder to seven class fire risk classification from aerial imagery. We introduce a reproducible partition of the public FireRisk training mirror and an implementation that records data provenance, preprocessing and model selection. Two initial runs compare a frozen encoder probe with full model adaptation. On the validation partition, full adaptation reaches 63.05% accuracy and 58.94% macro F1, compared with 55.95% and 50.19% for the probe. Both runs use one training seed and select their checkpoint on the same validation partition. These development results support further evaluation of SigLIP2 but do not establish performance on an independent test set or unseen regions. The accompanying code provides a common framework for repeated experiments and comparisons with additional visual encoders.

1 Introduction

Aerial images describe vegetation, buildings and surface cover that may help distinguish wildfire hazard categories. FireRisk links such imagery to labels derived from the Wildfire Hazard Potential map [4]. The task is to recover a hazard category assigned to an image tile. It does not directly measure the probability of a future fire at that location.

SigLIP2 learns visual representations using image and text supervision together with additional representation learning objectives [5]. This motivates an empirical question about whether its image encoder can adapt to the different visual content of aerial imagery. We study a small initial comparison and release its protocol and implementation for further experiments.

The distinction between recovering land cover cues and separating hazard levels matters in this task. Water and nonburnable surfaces may be visually distinct, while the five ordered hazard categories can share vegetation and built structures. A useful analysis must therefore examine class errors alongside aggregate accuracy. Our study reports both, using a frozen encoder probe and a fully adapted encoder under two declared training recipes.

2 Related work

FireRisk provides supervised and self supervised representation benchmarks for aerial fire hazard classification [4]. We examine transfer from a more recent pretrained image encoder under a separately declared partition.

Learning from aerial images also has a substantial dense prediction setting. Bıçakçı and Sarıca combined attention gates and a Transformer within a U Net for building segmentation from aerial imagery and laser data [1]. Their work illustrates the use of local and broader contextual representations in aerial scene interpretation. The present experiment instead uses RGB imagery alone and assigns one hazard category to each tile. Building masks and fire hazard labels are different targets, so segmentation performance does not imply hazard classification performance.

SigLIP trains image and text representations using a pairwise sigmoid objective [6]. SigLIP2 extends this family with captioning, self distillation, masked prediction and data curation objectives [5]. These encoders offer representations that can be reused beyond their original image and text training task.

In a geospatial application, Bıçakçı and colleagues used SigLIP image embeddings to retrieve gallery geolocations for multimodal language model prompts [2]. That study keeps the pretrained encoder for street imagery retrieval. Here we examine supervised adaptation of SigLIP2 to aerial hazard labels. We use only its image encoder and do not evaluate text prompts, language generation or geographic retrieval.

3 Data and protocol

The original FireRisk study describes 91,872 images, with 70,331 training examples and a separate validation partition [4]. Its RGB imagery was obtained from the National Agriculture Imagery Program during 2019 and 2020 and cropped to 320×320 pixels. Labels come from the 2020 Wildfire Hazard Potential map. We use the `blanchon/FireRisk` mirror, which exposes only the 70,331 training examples. The mirror therefore does not provide the published evaluation partition.

We create a new split with 49,231 training images, 10,552 validation images and 10,548 reserved test images. Allocation is stratified by class, using split seed 2026. The seven categories are very low, low, moderate, high, very high, nonburnable and water. The split preserves the class imbalance. For example, the validation partition contains 3,264 very low examples and 490 very high examples.

Very low accounts for 30.93% of the validation partition, compared with 4.64% for very high and 2.46% for water. Predicting the largest class for every image would therefore produce 30.93% validation accuracy. We retain the observed class frequencies instead of resampling the validation set. Macro F1 gives each category equal weight, while class recall exposes which categories are missed.

Before allocation, the preparation code hashes decoded RGB pixels and image dimensions using SHA256 after correcting image orientation. It groups exact duplicates and rejects conflicting labels within a group. The audit found 70,331 unique pixel groups and no repeated images in this revision. This check concerns exact identity. It cannot exclude overlapping tiles or visual similarity between nearby locations. The mirror contains image and label fields without coordinates or acquisition dates, so this experiment cannot establish geographic or temporal independence.

The data revision, model revision and split hash are recorded with each run. Validation macro F1 selects the checkpoint. The reserved test partition has not been evaluated in this study. As a result, our scores are development measurements under a new protocol and cannot be compared directly with the original FireRisk benchmark.

4 Model and training

4.1 Classifier and objective

We use the SigLIP2 base image encoder with 16×16 patches at 224×224 input resolution, distributed through timm. The pretrained attention pooling is retained. A trainable LayerNorm and a seven output linear layer operate on its 768 dimensional representation. The resulting classifier has 92,891,143 parameters. In the frozen encoder probe, only the head changes, leaving 6,919 trainable parameters. The encoder remains in evaluation mode during training. Full adaptation updates all parameters and uses head dropout with probability 0.1.

Let $f_\theta(x)$ denote the pooled image representation. At evaluation, the class probabilities are

$$p(y = c \mid x) = \text{softmax}(W \text{LN}(f_\theta(x)) + b)_c. \quad (1)$$

The frozen recipe updates W , b and the LayerNorm parameters while keeping θ fixed. Its head is therefore more flexible than a strictly linear map of raw encoder features. Both recipes minimize smoothed categorical cross entropy

$$\mathcal{L} = -\frac{1}{B} \sum_{i=1}^B \sum_{c=1}^K q_{ic} \log p(y = c \mid x_i), \quad q_{ic} = (1 - \varepsilon) \mathbf{1}[y_i = c] + \frac{\varepsilon}{K}, \quad (2)$$

where $K = 7$, B is the batch size and $\varepsilon = 0.05$. No image and text pairing objective is used during adaptation.

4.2 Preprocessing and optimization

Training uses random crops covering 80% to 100% of the image area, with aspect ratios from 0.9 to 1.1. Horizontal and vertical flips have probability 0.5, and rotations use multiples of 90 degrees. Evaluation uses deterministic bicubic resizing and the pretrained center crop policy with crop proportion 0.9. Each RGB channel is normalized with mean 0.5 and standard deviation 0.5.

Both runs use AdamW with weight decay 0.05, label smoothing 0.05 and gradient clipping at norm 1. Learning rates follow a cosine schedule after warmup over 10% of the configured optimization steps. The probe uses a head learning rate of 10^{-3} and at most 15 epochs. Full adaptation uses 10^{-5} for the encoder and 3×10^{-4} for the head, with at most 30 epochs. No class weighting or mixup is applied. The effective batch size is 64 in both cases. Full adaptation accumulates gradients over two batches of 32 images and uses gradient checkpointing. Computation uses bfloat16 on one NVIDIA RTX 5090 GPU. Training seed 42 is shared by the two runs.

Early stopping follows seven epochs without a validation macro F1 improvement greater than 0.0001. Both runs stop after 14 epochs and select epoch 7. The different learning rates, dropout and schedule lengths make this a comparison of two initial recipes. They prevent attribution of the observed difference to encoder adaptation alone.

5 Preliminary results

Table 1 reports uncalibrated predictions from the selected checkpoints. Full adaptation improves validation accuracy by 7.10 percentage points and macro F1 by 8.75 percentage points relative to the probe. Macro F1 assigns equal weight to the seven class F1 scores, making the minority classes visible alongside accuracy.

Recipe	Trainable parameters	Accuracy (%)	Macro F1 (%)	Time (min)	Memory (GiB)
Frozen encoder probe	6,919	55.95	50.19	12.1	0.74
Full adaptation	92,891,143	63.05	58.94	19.0	1.96

Table 1. Validation results from one training seed. Time covers the training loop, validation and checkpoint writing, excluding data preparation and model loading. Memory is peak PyTorch allocated GPU memory and does not represent total device or system memory.

5.1 Errors within hazard categories

Table 2 shows the validation support, F1 and recall for each class. Full adaptation has higher F1 for every category in these selected checkpoints, but recall does not improve uniformly. High recall changes from 46.56% to 45.08% even though its F1 increases. This is a useful distinction when missed hazard areas have different costs from false alarms.

Class	Support	F1 (%)		Recall (%)	
		Frozen	Full	Frozen	Full
Very low	3,264	60.86	69.87	59.56	68.96
Low	1,606	35.35	41.76	36.80	44.83
Moderate	1,293	29.40	42.03	25.91	42.00
High	945	40.27	46.53	46.56	45.08
Very high	490	21.29	39.90	14.49	33.47
Nonburnable	2,694	80.62	85.12	85.93	86.30
Water	260	83.53	87.33	80.00	86.15

Table 2. Class results from the same 10,552 validation images. Support is the number of true examples. Frozen denotes the frozen encoder probe and Full denotes full adaptation.

Full adaptation correctly identifies 164 of the 490 very high validation examples. Its 33.47% recall limits the usefulness of the current classifier when missing high hazard areas has a substantial cost.

Figure 1 shows that the remaining errors often fall into another hazard category. Under full adaptation, 29.70% of low examples are assigned very low and 26.88% of high examples are assigned moderate. Among

very high examples, 140 are predicted high and 110 moderate. Together these two destinations account for 51.02% of that class. Such errors can preserve a broad vegetation or surface interpretation while losing the intended hazard distinction. Establishing their physical cause would require image review and additional covariates.

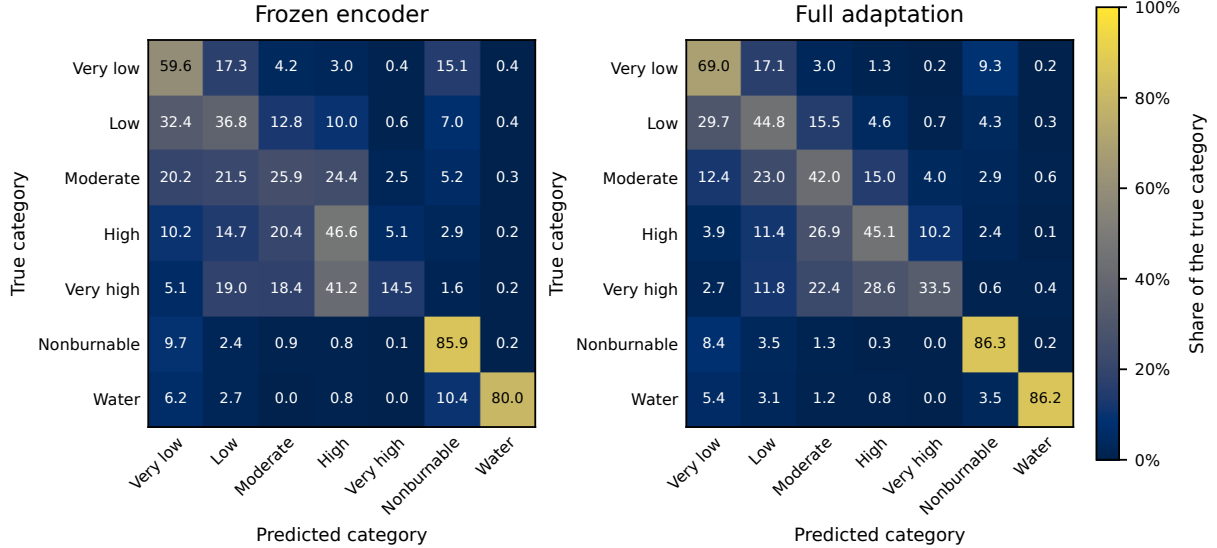


Figure 1. Validation confusion matrices normalized within each true class. Rows show true labels and columns show predictions. Every prediction is included, and each row sums to 100% before rounding.

The 7,598 images whose true labels belong to the five ordered hazard levels provide a more focused view. Exact category accuracy on these images rises from 44.50% to 54.01%. Predictions of water or nonburnable are retained as errors in this calculation. Of the 749 additional correct predictions across the complete validation set, 723 come from these hazard classes. The improvement is therefore not explained solely by easier water and nonburnable recognition.

The runs executed concurrently on separate RTX 5090 GPUs. Their timings are contextual measurements and do not provide a controlled speed comparison. Times exclude environment installation, downloads and the initial pixel audit. No repeated training seeds are available, so the variation caused by training randomness remains unknown.

6 Discussion

The observed result is consistent with a useful pretrained representation that benefits from task specific adaptation. It is also consistent with differences in head optimization and schedule. Our two recipes do not separate these explanations. A controlled follow up should match the search budget and optimization schedule before estimating the contribution of encoder updates. Independent seeds are needed to measure training variation.

The residual errors suggest a practical priority beyond increasing overall accuracy. Very high recall remains 33.47%, and confusion frequently moves examples toward lower hazard categories. Class weighting, alternative sampling or an ordinal objective are candidates for validation experiments. They may trade precision against recall and should not be presented as improvements without measured evidence. The current seven class loss treats all incorrect labels categorically and does not impose a distance between the five ordered levels. Water and nonburnable also require separate treatment rather than placement on that ordinal scale.

Geographic independence remains unresolved. Ploton and colleagues showed that ignoring spatial dependence can overstate predictive performance in ecological mapping [3]. That finding motivates caution here, without establishing the size of any bias in this mirror. Pixel deduplication cannot identify nearby or overlapping tiles. Coordinates and scene identifiers would be needed for a spatial holdout, and acquisition dates would support temporal evaluation. The model also learns map derived hazard labels from RGB

imagery. It has no weather, ignition or exposure inputs, so the present evidence does not justify using its output as a future fire probability or an operational warning.

7 Reproducibility

The implementation keeps original Parquet shards and can build an indexed archive of unchanged image bytes for shuffled access. It avoids extracting each image to a separate file and records the selected artifact location. Each run saves its configuration, environment, preprocessing, data manifest, training history and selected weights. Immutable dataset and model revisions and the split hash are listed in the accompanying manuscript documentation. The initial run artifacts do not record a source commit and mark the development tree as modified. The repository also includes presets for LoRA adaptation, DINOv2, ConvNeXt and ResNet50. Those presets have no completed comparative results here.

The code separates source files from an explicitly chosen artifact directory and provides a locked Python environment. Original shards occupy about 10.78 GiB, with a similar additional allocation for the optional image archive. A shard access mode avoids this second allocation when disk space is limited. The manuscript figures can be regenerated from the saved validation predictions using the supplied analysis script. Their source matrices and class metrics are included as compact numerical artifacts.

Final experiments should use a committed release and declared validation search budgets. The fixed recipes can then be evaluated on the reserved test partition. Possible overlap with the pretraining data remains unknown.

8 Conclusion

In one preliminary comparison on a fixed FireRisk mirror split, full SigLIP2 adaptation reaches 58.94% validation macro F1 against 50.19% for a frozen encoder probe. The gain extends to the five hazard categories, although very high recall remains limited and high recall slightly decreases. The protocol, code and class analysis provide a starting point for a broader study. Repeated controlled experiments and independent evaluation remain necessary before claiming reliable transfer to new regions.

Code availability github.com/yunusserhat/firerisk. The code is distributed under the GNU General Public License version 3. Upstream imagery and pretrained weights retain their respective terms.

References

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